

ARYA SHRESTHA

551-229-7377 | arya.shrestha1@marist.edu | linkedin.com/in/aryashrestha05 | github.com/AryaShrestha05

EDUCATION

Marist University

Bachelor of Science in Computer Science and Cybersecurity **GPA: 3.9**

Poughkeepsie, NY

Aug. 2024 – May 2028

- **Relevant Coursework:** Data Structures & Algorithms, OOP, Software Development I & II, Database Management, Discrete Mathematics, Data Communications

EXPERIENCE

Undergraduate Research Assistant

Marist University

Jan. 2026 – Present

Poughkeepsie, NY

- Designed a network protocol in **C** from scratch, cutting packet delay by **2.62x** for **7,000+** daily users on campus WiFi
- Automated a **24/7** testing pipeline across **6 TCP protocols** with **Python** and **Docker**, cutting packet loss by **65%**
- Wrote custom **Linux kernel modules** in **C** for zero downtime protocol benchmarking, driving a **20%** cut in retransmission
- Engineered a detection algorithm combining **3 network signals** to catch real failures while ignoring temporary WiFi noise

Software Engineer Intern

Advocacy Financial

May 2025 – Aug. 2025

Covington, LA

- Built a **legal SaaS platform** digitizing estate and succession workflows for **60+** attorneys, cutting prep time **40%**
- Designed data models and database architecture managing client asset portfolios across a **55-table** multi-tenant system
- Built a Jest-based automated testing suite covering **86** API endpoints, slashing production bugs by **40%**
- Designed a 3 tier client, attorney, superuser system by implementing **Node.js**, **Express**, **JWT**, achieving access control across **86** endpoints

PROJECTS

Room Foxes | **React Native, Expo, Node.js, Express, PostgreSQL, AWS RDS, Mixpanel, Jest**

- Built a roommate matching platform solving college housing compatibility, serving **200+** students across iOS and Android
- Engineered a compatibility algorithm across **12** lifestyle attributes with **80%** match satisfaction in user surveys
- Integrated Mixpanel tracking **15+** events per session, improving Day-7 retention by **35%**
- Optimized AWS-hosted PostgreSQL with connection pooling and indexed queries, cutting load time by **40%**

Docere | **Node.js, Supabase, Redis, Nginx, Docker, D3.js, Auth0, CI/CD, RAG**

- Built a RAG pipeline injecting lesson context per prompt, improving response relevance by **40%** for **100+** students
- Reduced AI response latency by **60%** via Redis caching and SQL-based learner progress tracking
- Deployed microservices via Docker, Nginx, and CI/CD pipelines, cutting deployment time by **50%** across **3** institutions
- Secured **\$10,000** in funding by presenting measurable learning outcome improvements across pilot schools in MN

ShipIt | **Node.js, TypeScript, Express, Gemini 2.5 API, GitHub API, Docker, SSE, Jest**

- Built a 5-phase AI agent eliminating the hours-long PR bottleneck, delivering full code-to-production in **under 90 seconds**
- Reduced PR turnaround from hours to a plain-English prompt via automated generation, builds, and deployment
- Closed the feedback loop between Jest test failures and model input, increasing agent task success rate by **40%**
- Streamed real-time agent status via SSE, improving developer observability across **50+** test deployments

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, Java, SQL

Frameworks: Node.js, Express.js, React, React Native, Next.js, Flask, PyTorch, OpenCV, MediaPipe

Developer Tools: Linux/Unix, Docker, Kubernetes, Nginx, Git & GitHub, AWS, Wireshark, CUDA C, Expo

Databases: PostgreSQL, MySQL, Redis, MongoDB

Concepts: REST APIs, GraphQL, OAuth 2.0, TCP/IP, CI/CD Pipelines, RAG, Agile/Scrum, Jest, Pytest

LEADERSHIP & INVOLVEMENTS

HeadStart Fellowship Tech Fellow (Spring 2026), **Propel2Excel Phase 2 Fellow** – 1% acceptance (2025–Present), **ColorStack Member** – Resume reviews and peer mentorship for freshmen, **Notion Campus Leader** – Marist Computer Science Society